

markeR Shiny Web App: User Tutorial

markeR is a web app for scoring RNA-seq samples against gene sets to quantify a phenotype of interest, such as senescence. It helps answer two kinds of questions: how well one or more gene sets mark a phenotype, assessed rigorously across scoring and enrichment methods (Benchmarking Mode), and which metadata variables are associated with a gene set (Discovery Mode). This tutorial walks through all tabs in the order you will typically use them: About, Data, Preprocessing, Gene Sets, Benchmarking Mode, and Discovery Mode.

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About Tab

Purpose: Confirm the app and package versions you are running, and find your way to the appropriate tab.

Why use this: Knowing the exact package version is important when comparing your results with those reported in the [markeR paper](#) or comparing analyses run at different times.

What you will see:

(A) The top-left corner shows the web app version.

(B) The warning message below the welcome message shows the version of the markeR R package on which the app was built.

(C) Tab summaries provide a brief explanation of what you can do in each tab. The underlying methodologies are described in greater detail in the [markeR paper](#).

(D) The Processing log shows the R version, machine, and package versions running in the background, plus every decision made during data loading and preprocessing.

The screenshot displays the 'markeR' web application interface. At the top, a navigation bar includes the 'markeR' logo with the version 'Web App 0.99.7', an 'About' tab, and other tabs: 'Data', 'Preprocessing', 'Gene Sets', 'Benchmarking Mode', and 'Discovery Mode'. A 'Report bug' link is on the right.

(A) The 'About' tab is active, showing a welcome message: 'Welcome to markeR! Have a collection of genes that putatively mark a phenotype, but not sure how to combine them and turn them into meaningful metrics for phenotype quantification? markeR helps you evaluate gene sets as phenotype markers and explore their information. Here's a quick guide to each section of the app:'.

(B) Below the welcome message, a warning box states: 'This app was built with the markeR Bioconductor package v1.2. For more on the methodology behind it, see our paper.'

(C) The 'Data' tab is selected, showing a summary of the data loading process. It includes sections for '1. Data', '2. Preprocessing', '3. Gene Sets', and '4. Benchmarking Mode', each with a brief description of the steps involved.

(D) The 'Processing log' is visible at the bottom, showing the R version, machine, and package versions running in the background, along with every decision made during data loading and preprocessing.

Data Tab

Load the three inputs required by markeR: gene expression data, sample metadata, and gene sets. All three must be present and correctly matched before the remaining tabs become available.

Data
Load your gene expression matrix, sample metadata, and gene sets. Everything downstream depends on what you configure here.

Data Preview

✓ Data ready: 38,537 genes × 545 samples
Proceed to Preprocessing to filter, normalise, and review your data. [→ Proceed to Preprocessing](#)

LOADED DATA SUMMARY

38,537 genes
545 expression samples
9 metadata rows × 10 vars
9 gene set(s)

Expression Preview

Show 5 entries Search:

	ERR1805188	ERR1805189	ERR1805190	ERR1805191	ERR1805196	ERR1805199	ERR1805200	ERR1805201	ERR1805202
A1BG	121.12	101.49	132.34	105.05	100.73	87.51	110.8	75.39	
A1BG-AS1	160.38	130.17	180.6	104.62	114.37	110.08	149.61	115.81	
A1CF	0	1	0	2	0	1	1	0	
A2M	919.33	859	772.97	1297.82	1537.86	321.79	396.95	367	
A2M-AS1	19.67	11	9	3.18	5.13	3.17	2	4	

Showing 1 to 5 of 38,537 entries Previous 1 2 3 4 5 ... 7708 Next

Metadata Preview

Show 5 entries Search:

sampleID	DatasetID	DatasetAccessionID	CellType	Condition	SenescentType	Treatment	Timepoints	Immortalised
----------	-----------	--------------------	----------	-----------	---------------	-----------	------------	--------------

Overview

Purpose: The three dropdown menus load the data markeR needs: an RNA-seq gene expression matrix, a metadata table linking each sample to clinical, phenotypic, or experimental variables, and the gene sets used to quantify the phenotype of interest.

What you will see:

- (A.1) Gene Expression Data, including the option to import data from GEO.
- (A.2) Metadata, see below.
- (A.3) Gene Sets, see below.

Gene Expression Data

Purpose: Load the RNA-seq expression matrix used in all downstream analyses.

Navigation:

1. In the sidebar, pick a data source.
 - a) **Example data:** the unprocessed senescence cohort used in the markeR paper. This is loaded by default.
 - b) **Processed example data:** the same cohort, preprocessed according to the methodology described in the paper.
 - c) **Upload your own file:**
 - Only **bulk RNA-seq** data is currently supported. scRNA-seq is not yet compatible with the underlying methods; support is planned for a future release.

- .csv / .txt: genes in rows and samples in columns; the first column contains gene names and the first row contains sample names. The first cell may be empty or "Gene". Comma, semicolon, and tab separators are detected automatically; decimals may use "." or ",". Trailing separators are ignored, and values should not be enclosed in quotation marks.
- .rds: a matrix or data frame with gene names as row names and sample names as column names.

d) **Import from GEO:** fetch expression data directly from a public GEO accession.

Importing from GEO (Beta)

Purpose: Retrieve expression data and metadata directly from a public GEO accession instead of uploading files.

Why use this: This can save you from manually downloading and reformatting supplementary files when you want to quickly explore or benchmark a public dataset.

What you will see (A.1):

- Accession field: enter the GEO accession code and click **Fetch from GEO**.
- Ranked supplementary files: if expression data is not detected automatically, supplementary files are ranked by an expression-likelihood score. This score reflects how closely each file matches the expected structure of RNA-seq expression data (genes in rows or columns and numeric values). Higher-ranked files are more likely to be suitable.
- Sample-matching tools: these appear when sample names do not match between expression columns and metadata rows. **Remove samples** removes samples missing from either side; **Match by keyword in a metadata column** (recommended) links samples using partial text overlap, for example "Library: sampleA" vs "counts.sampleA"; **Manually rename expression columns** lets you rename expression columns to match the metadata directly.

Good to know: GEO submissions are not standardised, so some accessions may not parse or import correctly. If an accession does not work as expected, please open an issue on [markeR's GitHub page](#).

Metadata

Purpose: Load the sample-level table containing the clinical, phenotypic, or experimental variables tested in downstream analyses.

Navigation:

1. Pick a metadata source.
 - a) **Example metadata:** matches the default example expression data.
 - b) **Upload your own:**
 - .txt / .csv / .tsv: one row per sample; the **first column must contain Sample IDs** matching the expression matrix column names. Remaining columns can contain any sample-level variables (group, condition, batch, etc.). The separator is detected automatically; a header row is required and quoted values are handled automatically.
 - .rds / .xlsx: a data frame with samples as rows; column names used as variable labels.
 - c) **From GEO:** if the expression data source is set to Import from GEO, the metadata associated with that accession is fetched and loaded automatically; this option is unavailable for any other expression data source.

Gene Sets

Purpose: Load the gene set(s) used to score your phenotype of interest throughout the other tabs.

Navigation:

1. Pick a gene set source.
 - a) **Example gene sets:** nine senescence-associated sets used in marker's paper.
 - b) **Upload your own:** see format rules below. You can select multiple files at once; each becomes its own gene set, collected into the same list.
 - c) **Paste gene set(s):** enter gene sets directly, with no file required. See format rules below.

Format rules:

- Providing the expected direction of change (+1/-1 per gene) for the phenotype is recommended because it allows more interpretable biological results than an undirected gene list.
- .csv / .txt / .tsv: one column of gene names (undirected), or two columns containing gene names and direction (directed). Do not include a header row; every line is read as data, starting from the first line. The separator (comma, semicolon, tab, or whitespace) is detected automatically. The file name (without extension) becomes the gene set name.
- .rds / .rda: a character vector of gene names, a two-column data frame (genes plus direction), or a named list combining either format.
- Paste gene set(s): provide a name for the gene set, then paste one gene per line into the text box. For a directional gene set, append the expected direction after a semicolon on *every* line, for example TP53;up and MKI67;down (+1/-1 are also accepted). For an undirected gene set, list genes with no semicolon on *any* line, for example ACTB. A single gene set cannot mix the two conventions: either every line specifies a direction, or none do. Click **Add another gene set** to enter additional sets, then click **Use these gene sets** to apply them.

Data Readiness Check

Purpose: Confirm that your three inputs are correctly matched before the remaining tabs become available.

What you will see:

(B) Green banner: sample names in the metadata correspond to the column names of the expression data, and the number of samples matches between the two. You are ready for the next steps.

(B.1) Yellow warning banner: shown at the top of the page if there is any mismatch between metadata and expression data (differing sample counts or non-matching sample names).

 **Sample mismatch** - 14 expression samples vs 12 metadata rows.
Use the sample management controls in the sidebar to fix this before proceeding.

If unresolved, access to the remaining tabs is blocked, since all three data types are essential to the analysis. The appropriate solution depends on the input source: for user-provided data, correct the files and upload them again; for GEO datasets, use the controls described above to resolve the mismatch.

Summary Panel

Purpose: Perform a final sanity check on the loaded data before starting the analyses.

What you will see:

(C) Browse sample names in the expression data and metadata, identify which metadata column likely holds Sample IDs, and click any gene set name to expand its full gene list. Also reports the total number of samples, genes, metadata variables, and gene sets loaded; check that these match your expectations before proceeding.

Preprocessing Tab

This tab operates on the gene expression data and metadata loaded in the Data tab; gene sets are not affected. It proceeds through five sequential steps. Each step becomes available after the previous one is completed or explicitly skipped, and completed steps collapse into a green summary card that you can reopen and redo at any time.

Why use this: Raw RNA-seq counts are not directly comparable across samples. Differences in sequencing depth, library complexity, lowly expressed genes, and technical batch effects can all masquerade as biological signal. This tab lets you inspect and, where appropriate, correct these issues before calculating phenotype scores, reducing the risk that apparent discriminatory power reflects technical artefacts rather than biology.

What you will see:

- (A) **Progress bar.** Sidebar progress tracker shows which step you are on, plus a **Reset all** button and a **Skip (use data as-is)** button that bypasses all five steps and sends your raw imported matrix straight through unmodified.
- (B) **Gene ID Conversion (banner).**
- (C) **The preprocessing step boxes.**

marker

Web App 0.99.5

[About](#)
[Data](#)
[Preprocessing](#)
[Gene Sets](#)
[Benchmarking Mode](#)
[Discovery Mode](#)

[Report bug](#)

(A)

Preprocessing

Normalize and quality-control your data before analysis. Steps are optional and applied in sequence.

A preprocessing report will be generated at the end, including all plots and any comments you add at each step.

Progress

- 0 Metadata Types
- 1 Sample QC
- 2 Gene Filtering
- 3 Normalisation
- 4 PCA & Batch

Current data:
 41,492 genes
 12 samples

[Reset all](#)

[Skip \(use data as-is\)](#)

(B)

Gene ID format detected: Ensembl (Human (Homo sapiens))

Your gene identifiers appear to be in **Ensembl (Human (Homo sapiens))** format. Would you like to convert them to gene symbols? Version numbers (e.g. .2 in ENSG00000141736.2) will be stripped automatically.

Conversion currently supports gene-level IDs only (e.g. Ensembl gene, RefSeq, Entrez). Transcript-level IDs (e.g. ENST..., ENSMUST...) will not be recognised or converted correctly - if your IDs are transcripts, keep the original IDs and handle the mapping to genes yourself beforehand.

Species:

Human (Homo sapiens)

[Convert to gene symbols](#)
[Keep original IDs](#)

(C)

Step 0 - Verify Metadata Column Types

Select the target type for each column. The table below previews the result. Columns highlighted in green will be converted; red columns cannot be converted without data loss and will be left unchanged.

SampleID	geo_accession	title	status	submission_date	last_update_date	type	channel_count	source_name_ch1	organism_ch1
count.HC24	GSM9797392	Healthy control, replic...	Public on Jun 14 2026	Jun 09 2026	Jun 14 2026	SRA	1	colon	Homo sapien
count.HC26	GSM9797393	Healthy control, replic...	Public on Jun 14 2026	Jun 09 2026	Jun 14 2026	SRA	1	colon	Homo sapien
count.HC27	GSM9797394	Healthy control, replic...	Public on Jun 14 2026	Jun 09 2026	Jun 14 2026	SRA	1	colon	Homo sapien
count.HC29	GSM9797395	Healthy control, replic...	Public on Jun 14 2026	Jun 09 2026	Jun 14 2026	SRA	1	colon	Homo sapien
count.HC36	GSM9797396	Healthy control, replic...	Public on Jun 14 2026	Jun 09 2026	Jun 14 2026	SRA	1	colon	Homo sapien
count.HC37	GSM9797397	Healthy control, replic...	Public on Jun 14 2026	Jun 09 2026	Jun 14 2026	SRA	1	colon	Homo sapien

Showing 6 of 12 rows

geo_accession

Character

title

Character

status

Character

submission_date

Character

last_update_date

Character

type

Character

channel_count

Character

source_name_ch1

organism_ch1

characteristics_ch1

molecule_ch1

extract_protocol_ch1

extract_protocol_ch1.1

taxid_ch1

Session log (47 actions - click to expand)

7

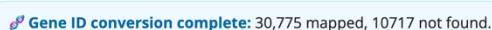
Gene ID Conversion (banner)

Purpose: Automatically detect whether your gene identifiers are Ensembl, RefSeq, or Entrez IDs and convert them to gene symbols.

Why use this: markeR's gene sets are defined using gene symbols. If your expression matrix uses Ensembl, RefSeq, or Entrez identifiers instead, conversion is required for the genes in your data to match the gene sets.

Navigation:

1. A banner appears above Step 0 if ID types other than gene symbols are detected.
2. If any gene IDs cannot be matched to gene names, the user is prompted to choose whether to discard the unmatched IDs or retain them in the analysis alongside the successfully matched gene names.

 Gene ID conversion complete: 30,775 mapped, 10717 not found.

10717 genes could not be mapped.

Choose what to do with them:

Good to know: Conversion supports gene-level identifiers only (for example, Ensembl gene, RefSeq, or Entrez IDs). Transcript-level identifiers (for example, ENST... or ENSMUST...) are not recognised or converted correctly. If your data uses transcript IDs, map them to gene symbols before uploading.

Good to know: If multiple original identifiers map to the same gene symbol, their counts are summed into a single row for that symbol. The mapping table (below the banner, after conversion) flags which identifiers were merged this way.

Step 0: Verify Metadata Column Types

Purpose: Confirm that each metadata column has the correct data type (character, numeric, integer, factor, or logical) before it is used in statistical tests.

Why use this: A numeric variable accidentally read as text (or vice versa) can change which statistical test markeR applies downstream – for example, correlation versus group comparison. Checking this here helps prevent misleading results.

Navigation:

1. Change any column's type via its dropdown. A live preview table updates immediately.
2. Check the preview table's highlighting.
 - a) **Green** means the conversion is safe.
 - b) **Red** means it would introduce missing values and is blocked.

Step 0 - Verify Metadata Column Types

Select the target type for each column. The table below previews the result. Columns highlighted in green will be converted; red columns cannot be converted without data loss and will be left unchanged.

SampleID	geo_accession	title	status	submission_date	last_update_date	type	channel_count	source_name_ch1	organism
count.HC24	GSM9797392	Healthy control, replic...	Public on Jun 14 2026	Jun 09 2026	Jun 14 2026	SRA	1	colon	Homo sa
count.HC26	GSM9797393	Healthy control, replic...	Public on Jun 14 2026	Jun 09 2026	Jun 14 2026	SRA	1	colon	Homo sa
count.HC27	GSM9797394	Healthy control, replic...	Public on Jun 14 2026	Jun 09 2026	Jun 14 2026	SRA	1	colon	Homo sa
count.HC29	GSM9797395	Healthy control, replic...	Public on Jun 14 2026	Jun 09 2026	Jun 14 2026	SRA	1	colon	Homo sa
count.HC36	GSM9797396	Healthy control, replic...	Public on Jun 14 2026	Jun 09 2026	Jun 14 2026	SRA	1	colon	Homo sa
count.HC37	GSM9797397	Healthy control, replic...	Public on Jun 14 2026	Jun 09 2026	Jun 14 2026	SRA	1	colon	Homo sa

Showing 6 of 12 rows

padding:6px 10px; margin-bottom:4px; font-size:0.8em; color:#991b1b; **Blocked conversions (would introduce NAs):**

- 'geo_accession': cannot convert character → numeric (would introduce NAs in 12 value(s))

geo_accession	title	status	submission_date	last_update_date	type	channel_count
Numeric	Character	Character	Character	Character	Character	Character

- Click **Apply changes** or **Confirm as-is** to proceed.

Step 1: Sample Complexity QC (optional)

Purpose: Identify low-complexity or under-sequenced samples before normalisation.

Why use this: A sample sequenced too shallowly, or with reads concentrated in very few genes, can distort downstream analyses. Identifying potential problems here makes them easier to investigate before interpreting later results.

Navigation:

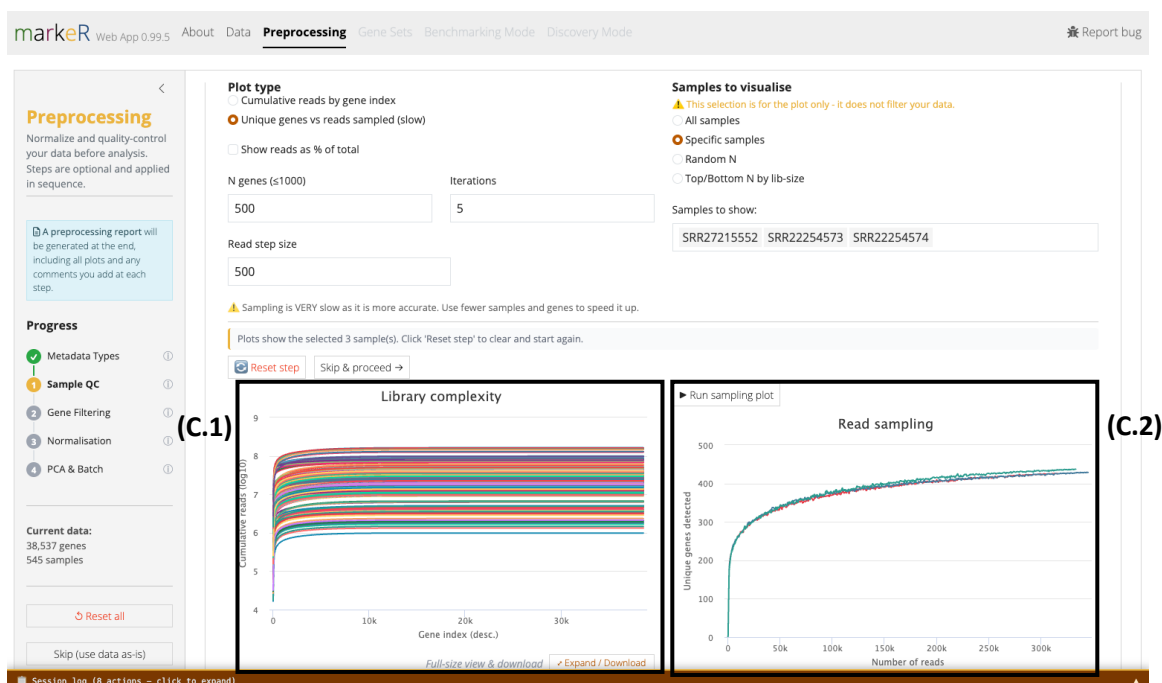
- Choose which samples to plot (all, specific, random N, or top/bottom N by library size).

What you will see:

(C.1) Library Complexity Plot: Cumulative reads by gene index (order by descending order of total number of counts): a per-sample line plot of cumulative reads sorted by rank (log₁₀ or percent of total).

(C.2) Read sampling plot: Unique genes vs. reads sampled: an iterative subsampling plot showing sequencing saturation and library complexity. Slower to compute; display-only, it does not filter your data.

Good to know: Both plots are interactive: hover over any line to identify the sample it corresponds to. If a sample looks suspect here, note it and remove it after Step 4 (see Removing Samples, below).



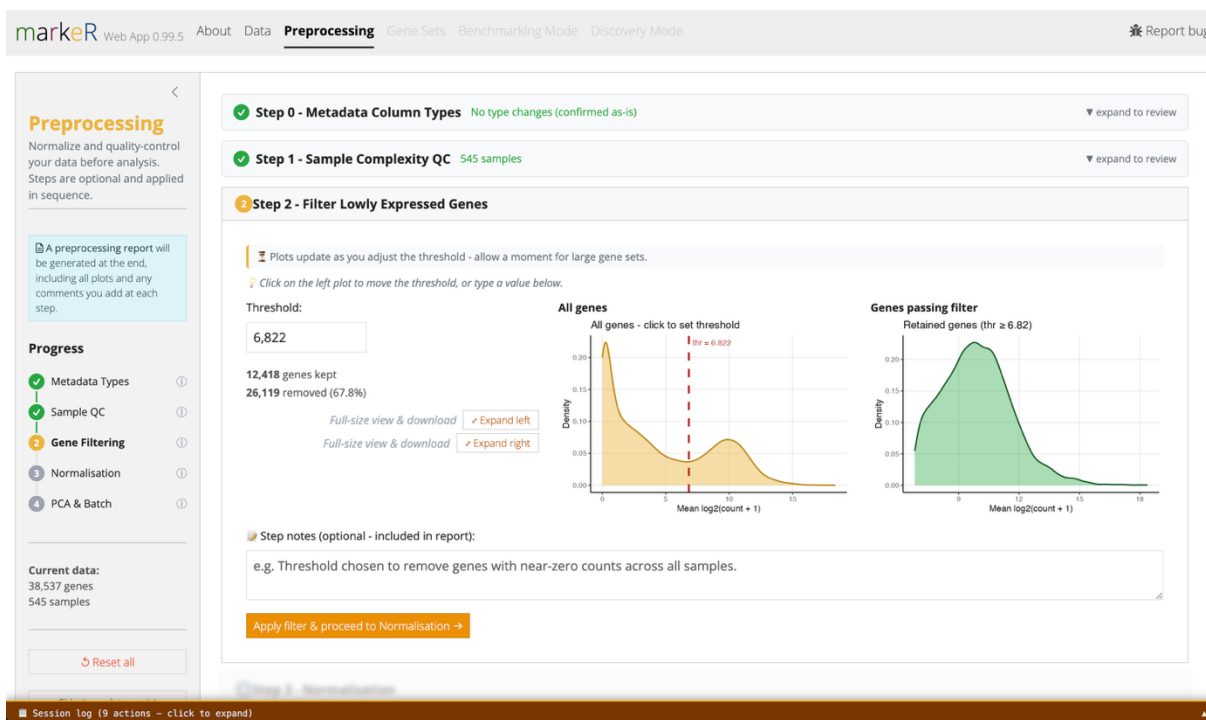
Step 2: Filter Lowly Expressed Genes

Purpose: Remove genes with expression levels too low for reliable measurement.

Why use this: Genes near the detection limit are sequenced too sparsely to distinguish real expression from chance; this sampling noise can add non-biological variability to downstream scores without contributing reliable signal.

Navigation:

1. A density plot shows all genes by mean $\log_2(\text{counts}+1)$. Click on the plot, or type a value, to set the threshold.
2. A second density plot and a live gene counter show how many genes pass the current threshold.

**Step 3: Normalisation & Quality Distribution**

Purpose: Put samples on a comparable expression scale before scoring.

Why use this: Without normalisation, a sample with more reads will tend to have higher raw counts across genes. A scoring method could mistake differences in sequencing depth for biological signal.

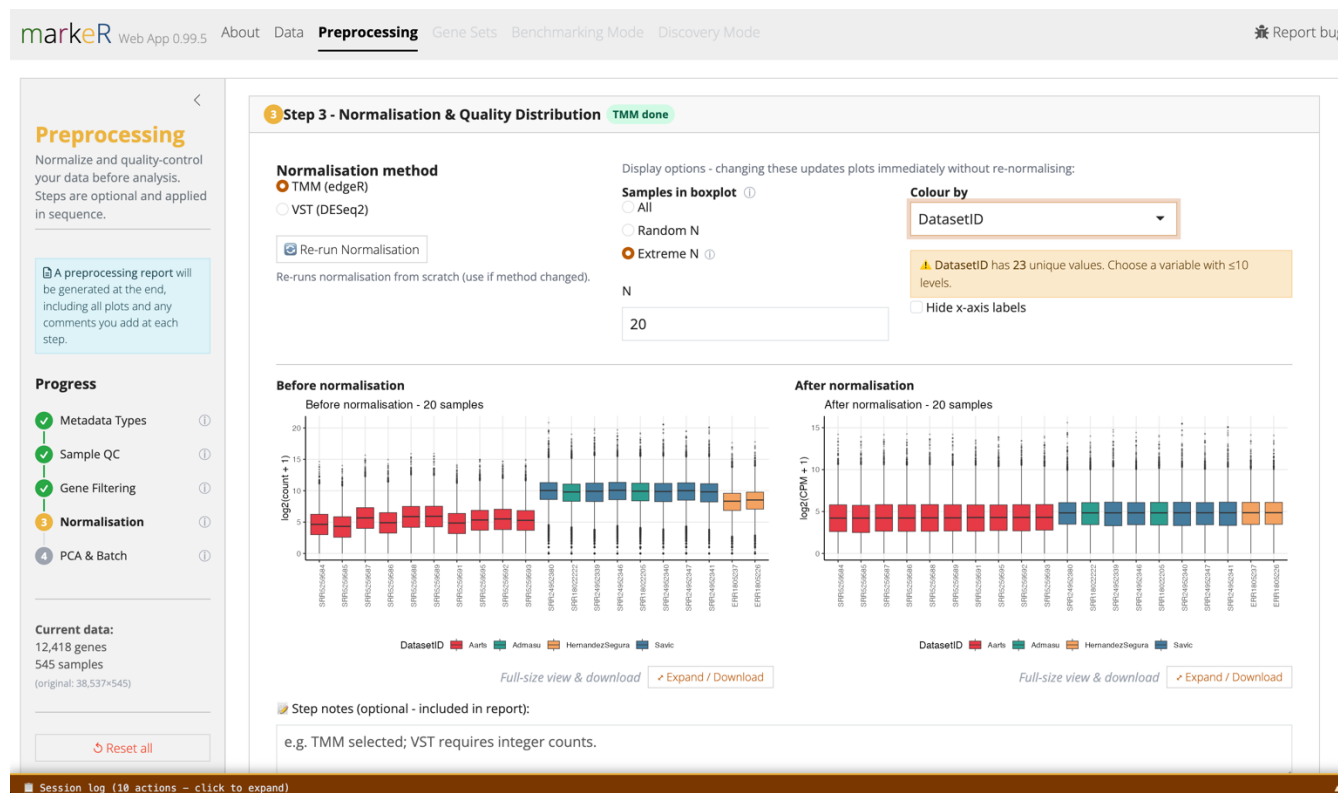
Navigation:

1. Choose a normalisation method.
 - a) **TMM** (edgeR calcNormFactors + cpm): the standard choice for most RNA-seq count data.
 - b) **VST** (DESeq2): requires non-negative integer counts. If your data look pre-normalised, the app warns but continues; if they look fractional, it rounds them automatically; negative values are blocked, with a pointer to use TMM instead.

What you will see:

(A) Before/after boxplots ($\log_2$ scale), fully reactive to display options (all, random-N, or extreme-N samples; colour-by metadata; hide x labels) without re-running normalisation.

Good to know: "Extreme N" shows the N/2 highest- and N/2 lowest-median samples together, which is useful for spotting outliers that a random sample might miss.



Step 4: PCA & Batch Correction (optional)

Purpose: Visualise sample structure and, if needed, remove known technical variation while preserving the biological variable of interest.

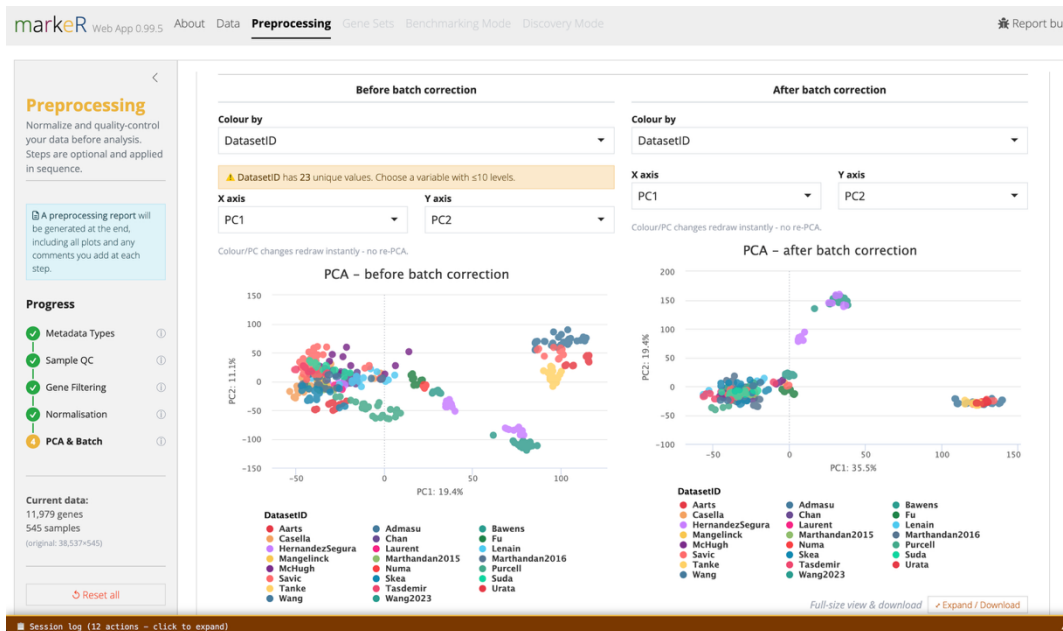
Why use this: If samples group by batch or processing date rather than by biology in the pre-correction PCA, gene set scores may reflect technical variation rather than the phenotype of interest.

Navigation:

- Run PCA on the normalised matrix (center is recommended and on by default; scale is off by default).
- For batch correction (limma *voom* + *lmFit*):
 - Variable(s) to remove (batch):** required, the technical variation to remove, for example DatasetID or processing batch.
 - Variable(s) to retain (signal):** optional, the biological variable you do not want removed, which may be partially confounded with the variable(s) to remove, for example Condition or Age.
- A second "after" PCA and scree plot let you visually assess whether the batch effect has been removed without collapsing your biological grouping.

What you will see:

(A) Both PCA plots are interactive: hover over a point to see its sample ID, group, and coordinates on the selected principal components.



Good to know: The batch-correction method is credited to voyAGER (eLife, 2024) and used in the marker paper (NAR Genomics and Bioinformatics, 2026).

Removing Samples

Purpose: Remove a sample identified as an outlier during Steps 1 through 4.

Navigation:

1. The sample-removal controls appear directly below the PCA plot at the end of Step 4.
2. Select the sample(s) to remove.
3. Optionally record a reason for removal, which is included in the preprocessing report.
4. Optionally attach any of the diagnostic plots already produced during Steps 1 through 4 (sample complexity, gene filtering, normalisation, or PCA) as supporting evidence; these are also included in the report.
5. Confirm the removal in the dialog that appears. Click **Remove selected & update data** to proceed.

Remove suspicious samples (only if needed)

Only use this if Steps 1-3 or the PCA plot above actually showed something suspicious about a sample. If everything looks fine, skip this and finalise directly below. Removing here updates the metadata to match and resets preprocessing back to Step 1, since gene filtering, normalisation and PCA were computed using these samples and must be redone for the reduced dataset.

Select samples to remove:

ERR1805188
ERR1805189
ERR1805190
ERR1805191
ERR1805196
ERR1805199

Attach supporting plot(s) as evidence in the report (optional):

☐ Gene filtering (Step 2)

☐ Normalisation (Step 3)

☐ PCA (Step 4)

Reason for removal (optional - included in report):

e.g. Outlier on PCA, separated from replicates of the same group.

[Remove selected & update data](#)

Good to know: Removing a sample resets preprocessing to Step 1 because gene filtering, normalisation, and PCA must be recalculated for the reduced dataset. After the first removal, the control collapses behind a "Remove more samples..." toggle.

Finalising Preprocessing

Purpose: Finalise your preprocessing choices and unlock the Gene Sets, Benchmarking Mode, and Discovery Mode tabs.

Navigation:

1. Choose how to finish.
 - a) If batch correction was run: **Use batch-corrected data** or **Use normalised only (no batch correction)**.
 - b) Otherwise: **Finalise & apply preprocessing**.

What you will see:

- (A) Preprocessing Summary card: lists genes kept/removed, samples removed, normalisation method, batch-correction status and variables, and the final data scale.

The screenshot displays the 'markerR' web application interface for the 'Preprocessing' step. The top navigation bar includes links for 'Web App 0.99.5', 'About', 'Data', 'Preprocessing' (active), 'Gene Sets', 'Benchmarking Mode', and 'Discovery Mode'. A 'Report bug' link is also present. The main content area shows a progress bar with five steps, all marked as complete. The 'Preprocessing Summary' card provides a detailed overview of the preprocessing steps and results. The sidebar on the left offers a 'Progress' overview and a 'Reset all' button. The bottom of the page shows a list of processed expression data samples.

- (B) Any "Step notes" you added along the way, plus every before/after plot, are compiled into an auto-generated HTML preprocessing report you can download for your records

Step notes (optional - included in report):

e.g. Batch corrected for sequencing run; retained condition effect.

- (C) Options to download the preprocessing report and data appear at the bottom left of the sidebar.

Good to know: Until preprocessing is finalised (or explicitly skipped), the Gene Sets, Benchmarking Mode, and Discovery Mode tabs stay locked with the tooltip "Complete preprocessing first."

The 'Download results' sidebar provides options for downloading data and reports. It includes buttons for 'Expression data (.csv)', 'Metadata (.csv)', 'Summary report (.html)', and 'Processing log (.txt)'. A section for 'Samples in report plots' allows users to select between 'All samples' and 'Extreme 20 (as in analysis)'.

Gene Sets Tab

Use this tab to assess whether your gene set(s) behave as expected at the individual-gene level, before performing the full analyses available in Benchmarking or Discovery Mode.

Good to know: Most sub-tabs operate on a single **Active gene set** chosen in the sidebar. **Pairwise Overlap** and **Similarity vs MSigDB** instead compare all loaded gene sets.

Pairwise Overlap

Purpose: Quantify the overlap between your loaded gene sets.

Why use this: Before interpreting multiple signatures as independent findings, it is useful to determine whether they capture essentially the same biology (high overlap, likely redundant) or distinct biological processes.

Navigation:

1. Choose a metric: **Jaccard index** or **Odds ratio**.
2. Click **Run Pairwise Overlap**.

Analysis deep dive:

- **Jaccard index** = $|A \cap B| / |A \cup B|$, ranging from 0 (no overlap) to 1 (identical sets).
- **Odds ratio** measures whether the observed overlap exceeds that expected by chance. A value of 1 indicates chance-level overlap, values >1 indicate enrichment, values <1 indicate depletion, and self-comparisons return Inf.

Similarity vs MSigDB

Purpose: Compare your gene sets against curated pathways from MSigDB or your own reference collections.

Why use this: A highly discriminatory gene set may simply recapitulate a well-characterised pathway (for example, a Hallmark gene set) rather than represent a novel biological finding. This analysis helps place your results into biological context and avoids over-interpreting novelty.

Navigation:

1. Choose a similarity metric (**Jaccard index** or **Odds ratio**), an organism (10 species supported), and an MSigDB collection (for example **Hallmark**) or subcollection (for example **CP:REACTOME**).
2. **Advanced Options:** restrict the analysis to selected pathways, define minimum similarity thresholds (or odds-ratio and adjusted p-value cut-offs), or upload/paste your own reference gene sets.
3. Click **Run Similarity**.

Correlation

Purpose: Examine how genes within the active gene set co-vary across samples.

Why use this: This analysis reveals whether the gene set behaves as a coordinated transcriptional programme or comprises several relatively independent expression patterns.

Gene Sets
Explore gene set composition, overlap with MSigDB, cross-set similarity, and per-gene expression profiles.
All analyses use only the genes from the loaded gene sets, not all genes in the dataset.

Active gene set
● Pairwise Overlap and Similarity use all loaded gene sets.

Gene set:
CellAge

Composition (1259 genes)
Search:

Gene	Direction
ABCA3	1
ACYP2	1
AKR1B1	1

Navigation:

1. Select the genes to display, choose a correlation method (**Spearman**, **Pearson**, or **Kendall**), and optionally split the heatmap by a metadata variable, generating one heatmap per group. This is useful when genes are expected to be co-regulated only under specific conditions.
2. Optionally cluster rows and columns and display labels, then click Plot Correlation Heatmap.

How to interpret this:

- Strongly correlated genes may be co-regulated or functionally redundant, whereas weakly correlated genes may contribute complementary information to the overall gene-set score.

Effect Size (Cohen's d)

Purpose: Rank genes according to how well each individually separates two groups.

Why use this: This analysis decomposes the overall gene-set signal into its individual gene contributions, helping identify genes that drive the phenotype and those contributing relatively little.

Navigation:

1. Select specific genes or automatically choose the Top N or Random N genes (ranked by |Cohen's d|, gene-set order, or randomly).
2. Select a grouping variable and its positive class, then click Plot Effect Size.

Analysis deep dive:

- Cohen's d measures the standardised difference in expression between the selected class and all remaining samples (one-vs-rest), using the pooled standard deviation.

How to interpret this:

- $d \approx 0$ indicates little or no separation; $d \approx 0.5$ a moderate effect; and $d \geq 0.8$ a large effect. Even when the selected variable has more than two levels, the comparison is always performed as one-versus-rest.

Gene PCA

Purpose: Assess whether expression of the genes in the active gene set alone captures the sample variation associated with a variable of interest.

Why use this: This analysis shows how much of the variation explained by the selected genes corresponds to the biological grouping of interest.

Navigation:

1. Choose a colouring variable and the principal components to display (PC1–PC10).
2. Click **Plot Gene PCA**.

Analysis deep dive:

- PCA is computed using only the expression matrix of the active gene set; all other genes are excluded.
- PCA is entirely unsupervised: the metadata variable used to colour the samples plays no role in calculating the principal components. Colouring is applied afterwards to assess whether the major axes of variation correspond to the biological grouping of interest.

Discriminatory Power (ROC / AUC)

Purpose: Quantify the ability of each individual gene to discriminate between two groups.

Why use this: This analysis summarises each gene's discriminatory performance using a single metric (AUC), allowing genes to be ranked according to their standalone predictive value independently of the overall gene-set score.

Navigation:

1. Select specific genes or automatically choose the **Top N** or **Random N** genes ranked by the effective $AUC = \max(AUC, 1 - AUC)$, which captures genes discriminating strongly in either direction.
2. Select a grouping variable and its positive class, then click **Plot Discriminatory Power**.

How to interpret this:

- An AUC of 0.5 indicates no discriminatory ability, whereas an AUC of 0 or 1 indicates perfect discrimination.

Good to know: An AUC below 0.5 does not imply that a gene is uninformative. Instead, it indicates that higher expression predicts the opposite class (for example, the gene is downregulated rather than upregulated in the condition of interest). Selecting **Invert AUC < 0.5 genes** simply displays these genes as $1 - AUC$, allowing all genes to be ranked on a common 0.5-1 scale irrespective of the direction of the association. The underlying ROC curve is also inverted.

Expression (Violins & Heatmap)

Purpose: Visually inspect expression patterns of individual genes.

Why use this: This provides a final qualitative check that genes exhibit the expected expression patterns before drawing conclusions from aggregate gene-set statistics.

Navigation:

1. **Violin plots:** Select genes, a grouping variable (x-axis), and a colouring variable, then click **Plot Violins**.
2. **Heatmap:** Optionally cluster rows and columns and display labels, then click **Plot Heatmap**.

Analysis deep dive:

- The heatmap displays row-scaled expression values (Z-scores computed independently for each gene), making expression patterns directly comparable across genes with widely different absolute expression levels.

Benchmarking Mode Tab

Purpose: Rigorously test and compare the ability of one or more gene sets to mark a phenotype using both score-based and enrichment-based methods.

Navigation:

1. In the sidebar, choose gene set scope: **All loaded** or **Select specific** sets.

Module A: Score Analysis

Shared settings for this module: pick a variable (or "None" for a density plot), choose a contrast mode for categorical variables or a correlation method for numeric variables, and optionally filter which contrasts are displayed.

Score Distributions

Purpose: Visualise per-sample scores across groups.

Navigation:

1. Choose a scoring method (Log-median, Ranking, or ssGSEA; see the *markeR* manuscript for further methodological detail) and a colour-by variable.
2. Optionally connect paired samples across groups, set a "focus" comparison for Cohen's d, and show Welch's t-test p-value in the subtitle.

Analysis deep dive:

- Log-median: $\log_2$ -transforms expression and median-centres each gene across samples.
- Ranking: non-parametric. Scores are based on each sample's within-sample rank of the gene set's members.
- ssGSEA: single-sample running-sum enrichment score (Barbie et al. 2009). Sensitive to gene set size.

Good to know: *Log-median calculates each score from centred, log-transformed expression and is therefore sensitive to normalisation and scale. Ranking uses each sample's within-sample ranks for the genes in the set rather than their expression magnitudes, making it more robust to outliers but discarding information about the magnitude of expression differences. ssGSEA computes a running-sum enrichment score and is sensitive to gene-set size. See the markeR paper for a detailed methodological comparison. Rather than relying on a single method, use All Methods Summary to assess whether results agree across methods.*

Classification (ROC / AUC)

Purpose: Test how well a gene set's score classifies a categorical variable.

Navigation:

1. Choose a scoring method and optionally check **Ignore direction** to fold AUC to 0.5 or above.

How to interpret this:

- An AUC of 0.5 indicates chance-level performance, whereas an AUC of 1.0 indicates perfect classification.

Good to know: *Off by default, direction is preserved: a curve can dip below the diagonal ($AUC < 0.5$), meaning the score runs opposite to how you defined the comparison, for example class order flipped, or it is really a "down" signature relative to this contrast. Checking "Ignore direction" folds any AUC below 0.5 to $1 - AUC$, so you can compare separation strength across many contrasts or gene sets without direction sign flipping the ranking, at the cost of hiding which way the score actually points. ROC curves appear also inverted.*

All Methods Summary

Purpose: Compare every gene set against every contrast, across all three scoring methods, in one view.

Why use this: Identifies which gene sets and contrasts are most consistently discriminatory regardless of scoring method.

Navigation:

1. Set significance and effect-size thresholds, then click to run.

Analysis deep dive:

- A Cohen's d/f heatmap (gene sets by contrasts, with cell text showing the effect size and adjusted p-value) plus an Effect Size Volcano (x-axis: effect size; y-axis: $-\log_{10}$ adjusted p-value; faceted by contrast).

How to interpret this:

- Dashed threshold lines mark your significance and effect-size cutoffs. Points beyond both lines, consistently across methods, are your best candidates.

Score FPR Simulation

Purpose: Test whether a gene set's discriminatory power is distinguishable from what would be expected by chance for random gene sets of the same size.

Why use this: A raw Cohen's d for a gene-set score does not tell you whether that particular gene combination is informative or whether a random set of the same size would produce a similarly large effect. The simulation provides an empirical null against which to assess score specificity.

Navigation:

1. Choose a contrast mode and number of simulations (10 to 2000; default 100). More simulations means longer runtime, since it scales with gene sets times simulations times 3 methods.
2. Click **Run Score FPR Simulation**.

Analysis deep dive:

- For each contrast, the app computes the observed Cohen's d/f for your gene set, then generates random gene sets of matching size from the full expression matrix to build a null distribution. A 95th-percentile threshold line marks where a random set would be considered "significant" purely by chance.

How to interpret this:

- If the observed effect size is above the null distribution's 95th percentile, the result is less likely to be explained by gene-set size alone. This tests score-level specificity, not gene-level discriminability; combine it with the per-gene diagnostics in the Gene Sets tab for a fuller assessment.

Score Results Table

Purpose: Export per-sample scores for further analysis.

Navigation:

1. Choose a method (or "All methods"), then download as CSV or Excel.

Module B: Enrichment Analysis (GSEA)

Purpose: Test for gene-set enrichment by performing differential expression analysis followed by GSEA, rather than relying on per-sample scores.

Why use this: Score-based methods reduce each gene set to one value per sample and ask whether that score discriminates between groups. GSEA first performs differential expression across the cohort and then asks whether the gene set is coordinately shifted in the contrast.

Navigation:

1. Choose a design mode.
 - a) **Automatic:** pick a grouping variable; the app builds the design matrix and contrasts for you.
 - b) **Manual:** paste an R formula, for example $\sim 0 + \text{Condition} + \text{Batch}$, for covariates or paired designs.
2. Set the ranking statistic (Auto, t-statistic, or B-statistic), number of permutations (default 1000), and p-adjustment settings, including whether to pool FDR correction across all contrasts.
3. Click **Run Enrichment Analysis (GSEA)**.

Good to know: *Ranking statistic: Auto (recommended) uses the t-statistic for directional gene sets (a data frame with +1/-1 gene weights) and the B-statistic for gene sets without direction. Use the t-statistic explicitly for a classic "up in disease, down in disease" signature, where preserving the sign of NES is biologically meaningful. Use the B-statistic for an unsigned gene list with no inherent direction, allowing both up- and down-regulated members to contribute to enrichment regardless of direction.*

DEG Volcanos

Purpose: Examine differential expression results for each contrast, with gene set membership highlighted.

Navigation:

1. Set logFC and adjusted p-value thresholds, and how many top genes to label.
2. Choose highlight mode: Gene set members or Significant genes by threshold.

How to interpret this:

- Grey indicates a background gene. Green indicates a gene annotated as up in the gene set. Red indicates a gene annotated as down. Dark blue indicates a gene set member with no assigned direction. Blue indicates a gene that is significant by your threshold.

Good to know: *Colour encodes each gene's annotated direction from the gene set itself, not the side of the plot on which the point falls for this particular contrast. For example, in a contrast A vs B where A is the positive class, a gene set annotated to mark class B will have its member genes coloured red (annotated down), even for genes that fall on the right-hand, up-logFC side of the plot for this contrast.*

Combined Volcano

Purpose: Compare NES and significance across all gene sets and contrasts.

Navigation:

1. This runs automatically once GSEA completes.

Analysis deep dive:

- Each point represents one gene set–contrast combination: the x-position reflects NES, the y-position reflects $-\log_{10}$ adjusted p-value, colour represents the gene set, and shape represents the contrast.

How to interpret this:

- Altered panel (undirected gene sets, B-statistic): a positive NES indicates enrichment among the most-altered genes.
- Enriched/Depleted panel (directed gene sets, t-statistic): a positive NES indicates enrichment among up-regulated genes, and a negative NES indicates enrichment among down-regulated genes.

NES Lollipop

Purpose: Rank gene sets by enrichment strength within each contrast.

Navigation:

1. This runs automatically once GSEA completes.

Analysis deep dive:

- Lollipop length reflects $|NES|$; point colour reflects adjusted p-value (red indicates significance). One panel is shown per contrast, split into Altered and Enriched/Depleted as described above.

Enrichment Plots

Purpose: Examine the classic GSEA running-enrichment-score curve for a specific gene set and contrast.

Navigation:

1. Select a gene set and contrast to inspect.

Analysis deep dive:

- The curve tracks the running enrichment score along the ranked gene list; bars along the bottom mark where gene set members fall in that ranking. A peak concentrated on one side indicates strong, coordinated enrichment.

Results Table

Purpose: Export full GSEA statistics for every gene set and contrast.

Analysis deep dive:

- Columns include pathway, NES, adjusted p-value, ranking statistic used, contrast, and leading edge.

Discovery Mode Tab

Purpose: Screen many metadata variables quickly against one gene set to generate hypotheses for further investigation.

Why use this: Quickly tests which clinical or technical variables (age, sex, batch, treatment, subtype, etc.) are associated with the gene set's signal before you invest time in rigorously benchmarking a specific variable. This mode is exploratory and assumes that the gene set is already a plausible marker of the phenotype of interest.

Navigation:

1. In the sidebar, pick one gene set.
2. Select the metadata variables to test. Variables with only one observed level are excluded automatically; variables with more than six levels are flagged as potentially unwieldy.
3. Choose a contrast mode: **Simple pairwise**, **One vs rest**, or **All combinations**. Hover over each option in the app for its exact definition and an example.
4. Set the significance threshold (adjusted p-value, default 0.05) and p-adjustment method.

Good to know: Discovery Mode is for hypothesis generation, not confirmation. If a result looks promising, examine it more rigorously in Benchmarking Mode and, regardless of how it performs there, seek confirmation in an independent dataset before relying on it.

Single Method Explorer

Purpose: Test one scoring or enrichment method against all selected variables.

Navigation:

1. Choose a method: Log-median score, Ranking score, ssGSEA score, or GSEA enrichment, then click **Run Selected Method**.

Analysis deep dive:

- Score methods output an Effect Size Overview (Cohen's f per variable), Score Distributions per variable, and Pairwise Contrasts (Cohen's d per pairwise comparison with adjusted p-values).
- The GSEA method outputs a lollipop plot: one row per contrast across all variables. Point position reflects NES (normalised enrichment score); colour reflects adjusted p-value significance.

How to interpret this:

- Dashed, grey points flag B-statistic contrasts with negative NES; these are generally uninformative, indicating no strong coordinated expression change. Rows below your significance threshold are highlighted in the results table.

Method Summary

Purpose: Compare all four methods across all selected variables in a single view.

Why use this: No single scoring or enrichment method is universally authoritative. A variable that shows a consistent signal across methods provides stronger evidence than one that is significant with only a single method.

Navigation:

1. Click to run. This is slower than Single Method Explorer, since GSEA runs once per variable.

Analysis deep dive:

- A combined heatmap: the left block shows Cohen's f per variable by scoring method (an asterisk marks p below your threshold); the right column shows the fraction of that variable's GSEA contrasts that were significant (for example, "2/3").